

Extrauterine Pregnancy 04. PUL: The Discipline of Not Knowing Yet

Companion reading handout for simultaneous listening.

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Transcript

Extrauterine Pregnancy audio course. Episode 4. PUL: The Discipline of Not Knowing Yet.

Pregnancy of unknown location is a temporary classification, not a diagnosis. That is the most important sentence in this lecture.

It should be heard twice because the phrase is deceptively neat. "Pregnancy of unknown location" sounds like a diagnosis because it has a name. But the name does not describe a disease; it describes a gap in information. The patient is pregnant, the ultrasound has not located the pregnancy, and the next step is a controlled process to resolve that uncertainty safely.

The term applies when the pregnancy test is positive and transvaginal ultrasound does not show a normally sited pregnancy or an ectopic pregnancy. It should be used only in a patient who is stable enough for follow-up. Barnhart's consensus statement was written because inconsistent language around this situation created clinical confusion. NICE makes the safety implication explicit: a woman with a pregnancy of unknown location could have an ectopic pregnancy until the location is determined.

PUL is therefore not a label that finishes the visit. It is a statement that the visit has not yet produced a location.

There are several possible final outcomes. The pregnancy may later be shown to be a normally sited intrauterine pregnancy. It may be an intrauterine pregnancy loss. It may be an ectopic pregnancy. It may resolve before the site is ever visualized, producing what the consensus language calls a failed or resolved pregnancy of unknown location. Sometimes the pregnancy remains persistent without location, and management must be individualized.

The clinical difficulty is that these outcomes can look similar at the first encounter. A stable patient may have mild bleeding, mild pain, a positive pregnancy test, and an empty uterus on ultrasound. That presentation does not immediately tell you whether the scan is simply too early, the pregnancy has already failed, or the ectopic pregnancy is still too small to see.

The resident's task is not to guess faster. The task is to make uncertainty safe.

NICE gives several rules that are worth hearing as clinical principles rather than guideline bullets. In PUL, symptoms deserve more weight than serum hCG results. If symptoms change, the patient must be reviewed regardless of previous hCG values or previous assessments. Two serum hCG measurements should be taken as close as possible to 48 hours apart, and not earlier, to guide subsequent management. Regardless of the serum hCG level, the patient needs written information about what to do if symptoms worsen and how to access emergency care 24 hours a day.

The 48-hour interval is not ceremonial. hCG needs enough time for a biologic trajectory to emerge. Too short an interval can turn random variation and assay noise into false certainty. Too long an interval can miss a patient who is becoming unsafe. Near 48 hours is the compromise that lets trophoblast behavior become interpretable while keeping the surveillance window clinically tight.

Those recommendations have one logic: hCG interpretation is useful only if the patient remains clinically safe while the next data point is being collected.

hCG turns a point into a trajectory. A single value confirms pregnancy and gives some context for ultrasound interpretation, but it does not locate the pregnancy. The second value, taken at the correct interval, begins to show whether trophoblastic activity is rising, falling, or plateauing. Even then, the curve is not the whole diagnosis. Ectopic pregnancies can rise, fall, or plateau. Some viable intrauterine pregnancies rise more slowly than older teaching allowed. A falling value may be reassuring only if symptoms are stable and follow-up continues to resolution.

So the listener should attach each hCG word to a limitation. Rising means active trophoblast, not location. Falling means declining trophoblast, not proof of intrauterine loss. Plateauing means abnormal behavior, not automatic permission for methotrexate. The curve tells you what the trophoblast is doing; ultrasound and clinical status tell you where danger may be and whether time is safe.

Progesterone should also be handled carefully. NICE recommends not using serum progesterone as an adjunct to diagnose either viable intrauterine pregnancy or ectopic pregnancy in women with PUL. Progesterone may contain information about viability in some research contexts, but it does not tell the resident where the pregnancy is implanted. A low value may fit a failing pregnancy; it does not safely exclude ectopic pregnancy.

The physiology explains the limitation. Early progesterone mainly reflects corpus luteum support of the pregnancy, and the corpus luteum is supported by hCG until placental steroid production takes over. That endocrine loop can operate whether trophoblast is in the endometrial cavity, the tube, the cervix, or a scar. A low progesterone may tell you that the hormonal support of the pregnancy is poor and that nonviability is more likely. It does not identify the tissue bed. Progesterone is therefore a viability-flavored biomarker, not a location test.

Prediction models such as M4 and M6 are useful because they try to formalize risk in stable PUL. M4 uses the hCG pattern from presentation and 48 hours later. M6 adds variables including initial hCG, hCG ratio, and progesterone, and later updates consider clinical factors such as age, pain, bleeding, and prior ectopic pregnancy. These models can reduce unnecessary visits and help sort low-risk from high-risk PUL, but they are not substitutes for clinical review.

Bobdiwala's clinical M4 study is instructive because it shows both the value and the boundary of modeling. The model classified many patients as low risk with high negative predictive value, but the study also documents that clinicians sometimes appropriately overruled the model and that unscheduled returns for pain, bleeding, or anxiety mattered. One serious adverse event occurred when two hCG values were taken 24 rather than 48 hours apart, the model was misused, and methotrexate was given to a pregnancy later found to be intrauterine. That case is exactly why timing and diagnosis discipline matter.

The model is not dangerous when used correctly in the right patient. The danger comes when the model is treated as authority detached from the clinical conditions that make it valid: stability, correct timing, correct input, access to follow-up, and readiness to reassess symptoms.

Sometimes the diagnostic question is not solved by waiting alone. Uterine aspiration can be used in selected PUL situations as a diagnostic tool, especially when the pregnancy is not desired or when a nonviable intrauterine pregnancy versus ectopic pregnancy must be separated before giving methotrexate. The logic is strict. If chorionic villi are identified in uterine tissue, the pregnancy was intrauterine. If no villi are found, or if hCG does not fall appropriately after aspiration, ectopic pregnancy remains on the table. Aspiration is therefore not a shortcut treatment for uncertainty. It answers the question only when tissue examination and post-procedure hCG behavior answer it.

This matters because uterine evacuation can falsely reassure. Removing tissue from the uterus does not prove that the pregnancy was in the uterus unless pregnancy tissue is actually identified or the hormone response fits complete intrauterine evacuation. The local termination position paper uses the same safety discipline before first-trimester termination: establish gestational-sac number, location, and gestational age when possible, and examine aspirated material for pregnancy tissue. In PUL, the resident should hear aspiration as a diagnostic maneuver with obligations, not as a way to make the uncertainty disappear.

A patient-centered explanation should be honest and structured.

You can say: "The pregnancy test is positive, but today's scan has not yet shown where the pregnancy is. There are several possible explanations. It may be too early to see a normally sited pregnancy, the pregnancy may be failing, or it may be ectopic. Because you are stable now, the safest next step is to repeat the hormone level at the correct interval and repeat the scan when it can answer the question. If pain worsens,

if you faint, if you develop shoulder-tip pain, or if you feel acutely unwell, you should not wait for the next appointment."

That explanation does not pretend certainty. It gives the patient a map.

It also gives the resident a map. While waiting, there are only three acceptable states: the patient is clinically stable with a defined follow-up plan; the patient has worsening symptoms and must be reassessed; or the diagnosis has become clear enough to move out of the PUL category. Anything else is drift. PUL management fails when the uncertainty is named but not actively managed.

PUL care also requires logistical honesty. A plan that is safe for a patient who can return in 48 hours, understands the warning symptoms, has phone access, and lives near emergency care may not be safe for a patient who cannot return, cannot be contacted, is leaving the country, or has escalating pain. Follow-up reliability is not paperwork. It is part of the clinical risk assessment.

This is why PUL belongs in early pregnancy services rather than in loose outpatient drift. Nature Reviews emphasizes that specialized early pregnancy assessment and timely ultrasound reduce delay. Farren notes that diagnostic uncertainty itself can produce psychological morbidity. The resident should not minimize that anxiety. The patient is being asked to live temporarily with the possibility of miscarriage, ectopic pregnancy, or a pregnancy too early to confirm. Clear timing, clear danger symptoms, and a named follow-up route reduce that burden.

The practical sequence is stable and repeatable.

First, decide whether the patient is stable. If she is unstable, severe, syncopal, peritonitic, or has concerning hemoperitoneum, PUL follow-up is not the primary plan.

Second, state what is known: positive pregnancy test, no pregnancy location yet on ultrasound.

Third, decide what the next data point is: usually repeat hCG near 48 hours, repeat ultrasound when indicated, and senior review if the pattern is concerning or unclear.

Fourth, make the interval safe: written warnings, 24-hour emergency access, and explicit instructions to return for worsening symptoms regardless of previous results.

PUL is the discipline of not forcing a diagnosis before the evidence is strong enough. It protects the patient from two opposite harms: missing an ectopic pregnancy during false reassurance, and exposing a potentially viable desired pregnancy to irreversible treatment before location and viability have been reasonably resolved.